

Leonardo Perez

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EDUCATION

Bachelor of Science in Mechanical Engineering

Expected: May/2028

The University of Texas at El Paso UTEP

Overall GPA: 3.92/4.00

Minor: Mathematics

Honors:

Dean's List: Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026

EXPERIENCE

Undergraduate Research Assistant

June/2025 – Present

W.M. Keck Center for 3D Innovation

El Paso, Texas

- Execute machine setup, powder sieving, and build preparation across multiple LPBF systems (EOS, SLM, Aconity, Renishaw, VELO3D) to ensure process-controlled conditions for each build, accumulating hands-on experience across 100+ processed samples.
- Machine components and custom fixtures for post-processing using a HAAS VF-2 CNC mill, programming toolpaths in Fusion 360 and SolidWorks to support internal and external LPBF research projects.
- Perform post-processing with Wire EDM (SSG HB400) to remove parts from build plates according to customer and research specifications, consistently meeting tolerances of ± 0.005 in.
- Operate, set up, and program a custom gantry system integrated into an RPMI 222XR using a Siemens Sinumerik controller, supporting Directed Energy Deposition (DED) research in a hybrid additive manufacturing environment.
- Develop and optimize CNC toolpaths in Siemens NX for hybrid additive manufacturing processes, ensuring accurate gantry motion and reliable deposition during experimental builds.
- Section DED specimens using Wire EDM (SSG HB400) to support metallurgical analysis, process validation, and characterization of hybrid manufactured components

Engineering Intern

June/2026 – August/2026

Vinton Steel, LLC

El Paso, Texas

- Designed and optimized a cooling pipe for hot rolling mill stands by developing and evaluating multiple concepts through ANSYS Fluent CFD, reducing spray deflection and improving bearing cooling with an estimated \$5,000/year maintenance savings.
- Authored a comprehensive Standard Operating Procedure (SOP) for roll stand assembly, standardizing maintenance practices and improving consistency, efficiency, and training for Roll Shop personnel.
- Installed and programmed NFC tags on rolling mill rolls, encoding identification and inventory data to improve roll traceability and support a more efficient inventory management system.

Manufacturing Team Lead

January/2026 – Present

Sun City Summit Rocket Team

El Paso, Texas

- Lead a manufacturing team producing flight-critical hardware for UTEP's participation at the International Rocket Engineering Competition (IREC), coordinating machining operations and task execution.
- Independently manufacture flight-critical components including rocket fins, fin mounting brackets, a steel nose tip, ballast, and bulk plates using Wire EDM and CNC milling, consistently achieving dimensional tolerances of ± 0.01 in.
- Perform dimensional inspection and quality verification of manufactured components to ensure precise fit, alignment, and aerodynamic performance prior to final integration.

ENGINEERING COMPETITIONS & ACHIEVEMENTS

First Place – America Makes Additive Manufacturing in Steel Competition

2026

Selected as one of two undergraduates on the multidisciplinary team that designed and manufactured an LPBF stainless-steel multi-tool; performed machine setup, build preparation, build removal, and Wire EDM post-processing for the competition-winning component.

Second Place – International Rocket Engineering Competition (IREC), 30k COTS Category

2026

Selected as one of 14 representatives from a 100+ member team based on technical contributions; supported final assembly, system integration, and launch operations at the competition

TECHNICAL SKILLS

- **Additive Manufacturing:** LPBF systems (EOS, SLM 280, Renishaw, Aconity, VELO3D), powder handling & sieving, build setup, post-processing
- **Subtractive Manufacturing:** CNC machining (milling & turning), Wire EDM, manual mill & lathe, bandsaw
- **Software:** Fusion 360 & SolidWorks (CAD/CAM), Ansys, MATLAB, Siemens NX, Microsoft Office.
- **Languages:** Fluent in English and Spanish